

RSA key generation, encryption, and decryption example.

Step 1: Choose two prime numbers p and q , and compute N and Euler's totient function $\phi(N)$:

$$p = 7, \quad q = 17$$

$$N = p \cdot q = 119$$

$$\phi(N) = \phi(p \cdot q) = (p - 1) \cdot (q - 1) = 6 \cdot 16 = 96$$

Step 2: Choose the value of e such that:

$$\gcd(e, \phi(N)) = 1 \quad \text{and} \quad 1 < e < \phi(N)$$

For example:

$$e = 17$$

Bob's public key will be equal to:

$$p_k^{bob} = (N, e) = (119, 17)$$

Step 3: To compute Bob's private key, we need to find d , such that:

$$d \cdot e = 1 \quad \text{mod } (\phi(N))$$

The value of d can be computed using the Extended Euclidean Algorithm:

$$r_0 = a = \phi(N) = 96$$

$$r_1 = b = e = 17$$

$$r_2 = r_0 - q_1 \cdot r_1 = a - 5 \cdot b = 96 - 5 \cdot 17 = 11$$

$$r_3 = r_1 - q_2 \cdot r_2 = b - 1 \cdot (a - 5b) = -a + 6b = 6$$

$$r_4 = r_2 - q_3 \cdot r_3 = (a - 5b) - 1 \cdot (-a + 6b) = 2a - 11b = 5$$

$$r_5 = r_3 - q_4 \cdot r_4 = (-a + 6b) - 1 \cdot (2a - 11b) = -3a + 17b = 1$$

Thus:

$$d = 17 = e$$

Although $d = 17 = e$ satisfies the RSA congruence ($ed \equiv 1 \pmod{\phi(n)}$), using identical values for the public and private exponents is not valid in secure RSA implementations for the following reason: RSA is designed as an asymmetric cryptosystem, where e is public and d is private. If $e = d$, then anyone with the public key can decrypt the ciphertexts, violating the confidentiality requirement. We thus need to choose a different value for e and recompute d .

Step 2 (again): Choose the value of e , for example:

$$e = 13$$

Bob's public key will be equal to:

$$p_k^{bob} = (N, e) = (119, 13)$$

Step 3 (again): To compute Bob's private key, we need to find d , such that:

$$d \cdot e = 1 \quad \text{mod } (\phi(N))$$

The value of d can be computed using the Extended Euclidean Algorithm:

$$\begin{aligned} r_0 &= a = \phi(N) = 96 \\ r_1 &= b = e = 13 \\ r_2 &= r_0 - q_1 \cdot r_1 = a - 7b = 96 - 7 \cdot 13 = 5 \\ r_3 &= r_1 - q_2 \cdot r_2 = b - 2(a - 7b) = b - 2a + 14b = -2a + 15b = 3 \\ r_4 &= r_2 - q_3 \cdot r_3 = (a - 7b) - 1(-2a + 15b) = a - 7b + 2a - 15b = 3a - 22b = 2 \\ r_5 &= r_3 - q_4 \cdot r_4 = (-2a + 15b) - 1(3a - 22b) = -2a + 15b - 3a + 22b = -5a + 37b = 1 \end{aligned}$$

Thus:

$$d = 37$$

Bob's private key will be equal to:

$$s_k^{bob} = (d, p, q) = (37, 7, 17)$$

Encryption and decryption steps:

Let us assume that we want to encrypt a plaintext $P = 10$.

Encryption process using Bob's public key:

$$\begin{aligned} C &= P^e \quad \text{mod } (N) \\ &= 10^{13} \quad \text{mod } (119) = 45 \end{aligned}$$

Decryption process using Bob's private key:

$$\begin{aligned} P' &= C^d \quad \text{mod } (N) \\ &= 45^{37} \quad \text{mod } (119) = 10 = P \end{aligned}$$